AHARADA EDUCATION

Excellence in Aviation, Aerospace & Advanced Computing Technologies

OFFICIAL CURRICULUM STUDY MATERIAL & LECTURE NOTES

COURSE & SUBJECT SPECIFICATION

Course: MBA Data Analytics & Artificial Intelligence
Subject Code: MBA - AI105
Subject Name: Data Base Management & SQL
Semester: Semester 1
Target Unit: Unit 2
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Unit 2: Data Base Management & SQL

Subject: Data Base Management & SQL (MBA - AI105)

1. Unit Syllabus & Module Breakdown

Entity Relationship (ER) Model; ER Diagrams; Entities, Attributes, Relationships; Cardinality; Keys (Primary, Foreign, Composite, Candidate); Relational Schema; Functional Dependencies; Database Design Process; Normalization (1NF, 2NF, 3NF, BCNF); Denormalization; Data Integrity Constraints; Business database modelling.

2. Core Learning Objectives & Key Concepts

- **Topic 2.1:** Entity Relationship (ER) Model
- **Topic 2.2:** ER Diagrams
- **Topic 2.3:** Entities, Attributes, Relationships
- **Topic 2.4:** Cardinality
- **Topic 2.5:** Keys (Primary, Foreign, Composite, Candidate)
- **Topic 2.6:** Relational Schema
- **Topic 2.7:** Functional Dependencies
- **Topic 2.8:** Database Design Process
- **Topic 2.9:** Normalization (1NF, 2NF, 3NF, BCNF)
- **Topic 2.10:** Denormalization
- **Topic 2.11:** Data Integrity Constraints
- **Topic 2.12:** Business database modelling.

3. Detailed Chapter Notes & Concepts

3.1 Entity-Relationship (ER) Modeling & Diagrams

The ER model defines the conceptual view of a database. Key components include Entities (real-world objects with independent existence), Attributes (properties or characteristics of entities such as simple, composite, single-valued, multi-valued, and derived), and Relationships (associations among entities). Cardinality ratios define the number of relationship instances (1:1, 1:N, M:N). Participation constraints specify mandatory or optional existence.

3.2 Relational Schema & Relational Integrity Constraints

A relational database consists of tables (relations) with rows (tuples) and columns (attributes). Integrity constraints maintain data validity:

• Entity Integrity: The primary key cannot contain null values.
• Referential Integrity: A foreign key value must match an existing primary key or be null.
• Domain Integrity: Column values must adhere to defined data types and ranges.

3.3 Normalization & Functional Dependencies

Normalization minimizes data redundancy and avoids insertion, update, and deletion anomalies:

• 1NF (First Normal Form): Eliminates repeating groups; all attribute values are atomic.

• 2NF (Second Normal Form): In 1NF and contains no partial functional dependencies (every non-prime attribute is fully functionally dependent on the entire primary key).

• 3NF (Third Normal Form): In 2NF and contains no transitive functional dependencies (X -> Y and Y -> Z where Z is non-prime).

• BCNF (Boyce-Codd Normal Form): Stricter version of 3NF where for every functional dependency X -> Y, X must be a super key.

4. Review Questions & Self-Assessment

1. Explain the fundamental architectural distinctions between physical and logical data independence.
2. Discuss how the topics covered in this unit apply to industrial workflows in MBA Data Analytics & Artificial Intelligence.
3. Formulate standard case diagrams and explain key constraints discussed under Unit 2: Data Base Management & SQL.
4. Review previous university examination questions related to Data Base Management & SQL (MBA - AI105).